

The background features a dark blue gradient with large, overlapping, curved shapes in shades of orange, pink, and purple on the right side, creating a modern, abstract design.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

SVS203-R

Thinking serverless

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About me



James Beswick

- Leading AWS Serverless Developer Advocacy
- Serverless geek
- Previously
 - Software Engineer and Product Manager
 - Tech guy for multiple startups and Fortune 500s
 - AWS customer since 2012

Agenda

Where we have come from – servers

Where to start with serverless

Good practices of serverless design

Whiteboarding

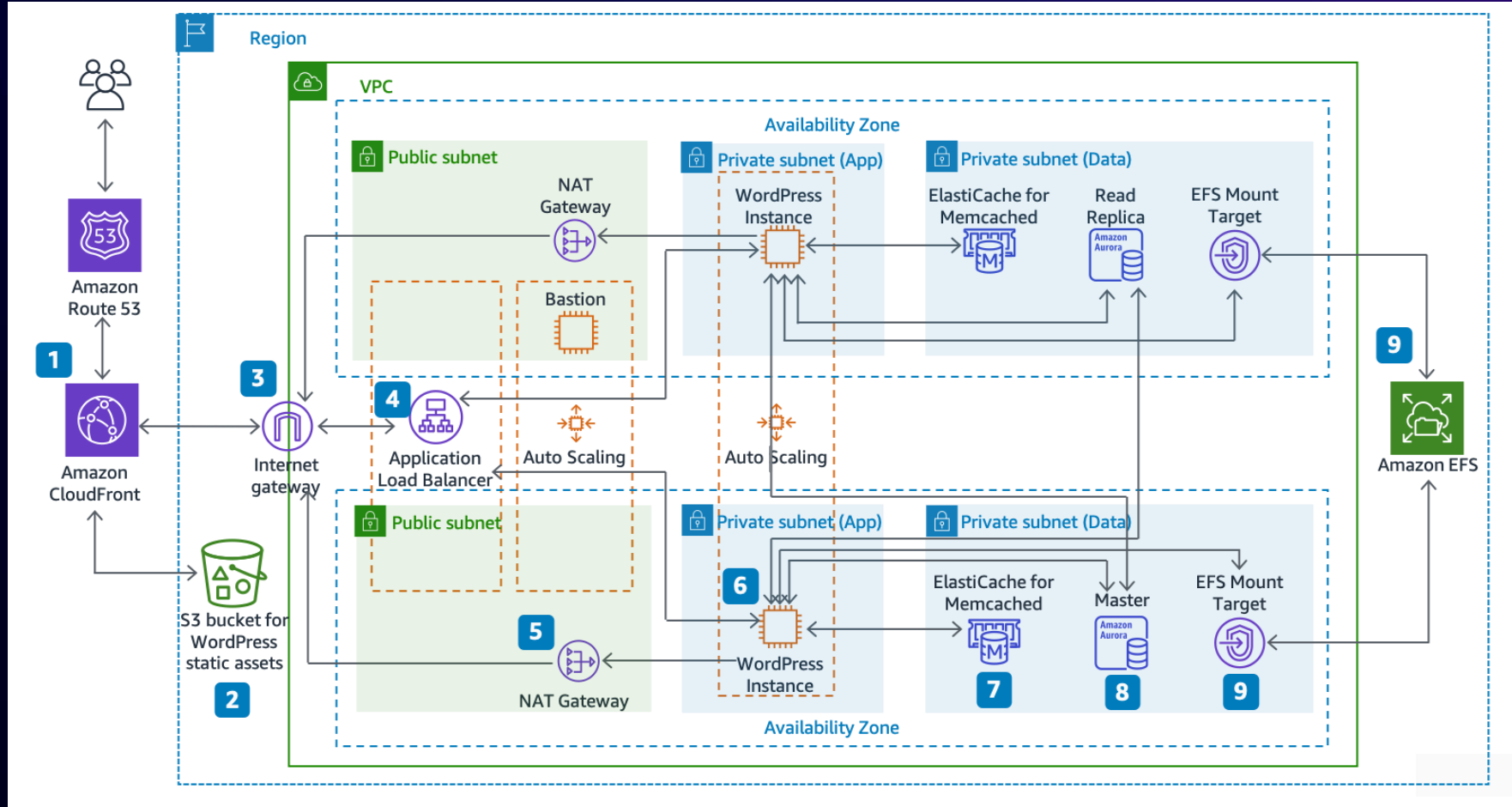
Servers



How do we use servers?

- State management
- Monolithic container for functionality
- One version, one server
- Server is an atomic unit of thinking
- The challenges of this model

For example: WordPress hosting on AWS



Source: <https://docs.aws.amazon.com/whitepapers/latest/best-practices-wordpress/reference-architecture.html>

Where to start with serverless



How AWS Lambda fits in

Attributes

- Runs on demand
- Supports many runtimes
- Responds to events
- Stateless
- Automatically scales

Best practices

- Avoid lifting and shifting
- 1 Lambda function per purpose
- Keep functions small
- Choose the right runtime
- Use functions for business logic and plumbing between services
- Include security

General approach to thinking serverlessly



Features first



Focus on events



Statelessness



Data flow



Use the services

What are serverless services?



Amazon
EventBridge



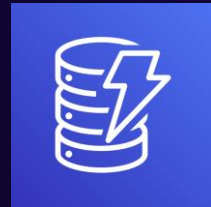
Amazon SNS



Amazon SQS



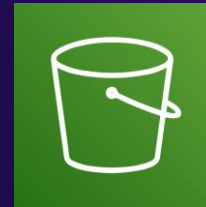
Amazon API
Gateway



Amazon
DynamoDB



AWS
Lambda



Amazon S3



AWS Step
Functions



Amazon
Kinesis



AWS
IoT Core



Amazon Elastic
Transcoder

Integrating with other AWS services



Amazon
Rekognition



Amazon
Comprehend



Amazon
Textract



Amazon
Transcribe



Amazon
Translate

Introducing ...



AWS Serverless Application Model (AWS SAM)

- AWS CloudFormation extension optimized for serverless
- Serverless resource types: Functions, APIs, tables
- Supports anything CloudFormation supports
- Open specification (Apache 2.0)



To learn more, visit:

<https://github.com/awslabs/serverless-application-model>

Example AWS SAM template

```
AWSTemplateFormatVersion: '2010-09-09'  
Transform: AWS::Serverless-2016-10-31  
Description: patterns-translation
```

```
Parameters:  
  TargetLanguage:  
    Type: String  
    Default: 'fr es it'
```

```
Resources:  
  TranslationBucket:  
    Type: AWS::S3::Bucket
```

```
  TranslatorFunction:  
    Type: AWS::Serverless::Function  
    Properties:  
      CodeUri: translatorFunction/  
      Handler: app.handler  
      Runtime: nodejs18.x  
      MemorySize: 128
```

```
  Environment:  
    Variables:  
      TargetLanguage: !Ref  
      TargetLanguage  
    Policies:  
      - S3CrudPolicy:  
        BucketName: !Ref  
      ResultsBucketName  
    Events:  
      OriginalTextUpload:  
        Type: S3  
        Properties:  
          Bucket: !Ref  
      TranslationBucket  
        Events: s3:ObjectCreated:*  
        Filter:  
          S3Key:  
            Rules:  
              - Name: suffix  
                Value: '.txt'
```

Transforms YAML into infrastructure

```
AWSTemplateFormatVersion: '2010-09-09'  
Transform: AWS::Serverless-2016-10-31  
Description: patterns-translation
```

Parameters:

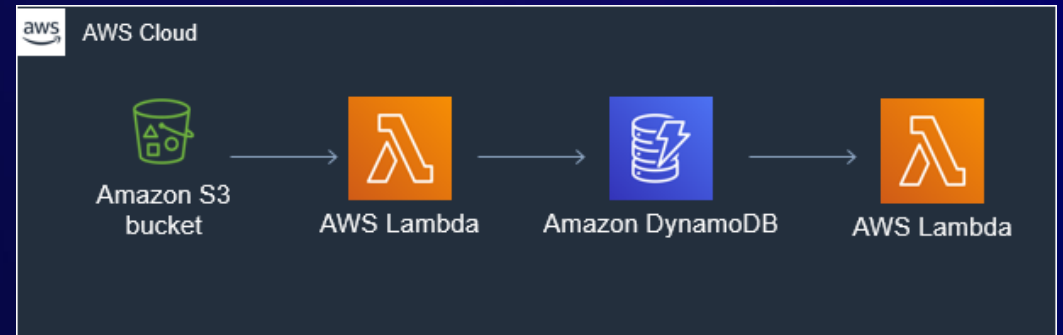
```
TargetLanguage:  
  Type: String  
  Default: 'fr es it'
```

Resources:

```
TranslationBucket:  
  Type: AWS::S3::Bucket
```

TranslatorFunction:

```
Type: AWS::Serverless::Function  
Properties:  
  CodeUri: translatorFunction/  
  Handler: app.handler  
  Runtime: nodejs18.x  
  MemorySize: 128
```



Whiteboarding



1. Form upload

Create a serverless application to support a customer review form submitted from a webpage

Incoming responses must be translated into English

Allow user to upload image with a response

Email any negative comments immediately

Only allow signed-in users to post reviews

2. Restaurant reservations

Create a serverless application to allow restaurant customers to reserve a table by SMS text message

Add a display showing upcoming reservations in real time

Speak the name of a guest when ready to seat table

Send a daily reservation email report

Alert a legacy application when reservations are made

3. Video streaming application

Create a serverless video streaming application, allowing broadcasting of live content

Allow on-demand
replay of
videos

Create a
transcription
for each video

Create a title
and summary
of each video

Enable fast-
forward/
rewind GIFs



ServerlessVideo

A live video streaming application built
with serverless technologies



ServerlessVideo

<https://s12d.com/video>

Wrap-up



Content
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Filters (3 resources)

Skill Level

☐ Novice
☐ Intermediate
☐ Advanced

Resource type

☒ Guides
☐ Workshops

Category

☐ AI/ML
☐ API
☐ Compliance
☐ Cost
☐ Data processing
☐ Event-driven architecture
☐ Infrastructure as Code
☐ Migration
☒ Operations

Learn from a collection of resources around common topics, including blog posts, videos, workshops and other training. Choose a learning path from the list below to learn more.

AWS Lambda Operator Guide

This guide is built for developers and operators of Lambda-based applications. It is aimed at operators of typical production serverless applications, looking to understand more clearly how to build, measure, troubleshoot, and optimize their compute processes. This guide covers concepts and best practices for designing Lambda-based applications, together with an approach for ongoing monitoring and troubleshooting.

View guide →

Implementing governance in depth for serverless applications

by Heeki Park , Pallavi Srivastava , Debasis Rath

Intermediate

35 minutes

View guide →

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Serverless Patterns Collection

Use serverless patterns to quickly build in production. Filter by pattern and copy the template directly into your application.

Application Load Balancer
Amazon Cognito
AWS Lambda

ALB, Amazon Cognito and Lambda integration

Create an ALB, Cognito and Lambda integration to fetch regional metrics

SAM

Python

View pattern

Application Load Balancer
AWS Lambda

Application Load Balancer with Lambda as a target

Create an Application Load Balancer with Lambda as target using CDK

CDK

TypeScript

View pattern

For more serverless learning resources, visit:
<https://serverlessland.com>

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Thank you!

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